

Foot MRI — 30 Sample Cases

A teaching file of 30 commonly encountered foot MRI diagnoses spanning the forefoot, midfoot, hindfoot, and marrow/infection processes, each with a realistic worked case.

Forefoot

Case 1 — Second MTP plantar plate tear

Clinical: 54-year-old woman, progressive forefoot pain and a "walking on a pebble" sensation, with early second toe drift.

F: On the sagittal and short-axis fat-suppressed PD images, there is a full-thickness tear of the **plantar plate** at its distal insertion onto the base of the second proximal phalanx, with a 3 mm gap and proximal retraction of the plate. Fluid signal fills the defect and tracks into the **second MTP joint**, which shows a small effusion. The **collateral ligaments** are intact. There is mild dorsal subluxation of the proximal phalanx and adjacent flexor tendon sheath fluid. No marrow edema in the metatarsal head.

I: Full-thickness distal tear of the second MTP plantar plate with associated joint effusion and early dorsal subluxation.

Case 2 — Morton neuroma, third webspace

Clinical: 46-year-old woman who wears narrow heels, burning interdigital pain relieved by shoe removal, with positive Mulder click.

F: In the **third intermetatarsal space**, there is a teardrop-shaped mass measuring 7 × 5 mm that is plantar to the intermetatarsal ligament. It is isointense to muscle on T1 and intermediate-to-low signal on T2, with mild homogeneous enhancement. No cystic component. The mass demonstrates the characteristic dumbbell morphology splaying the third and fourth metatarsal heads. No adjacent fluid collection greater than 3 mm to suggest bursitis.

I: Third-webspace Morton neuroma (7 mm), typical low T2 signal and morphology.

Case 3 — Intermetatarsal bursitis

Clinical: 39-year-old runner, diffuse forefoot ache worse with activity, no discrete trigger point.

F: Within the **second intermetatarsal space**, dorsal to the intermetatarsal ligament, there is a well-defined fluid-signal collection measuring 6 mm in transverse dimension, bright on fat-suppressed T2 and following fluid on all sequences with thin peripheral enhancement. No solid nodular tissue to suggest neuroma. The adjacent **plantar plates** and flexor tendons are intact. Minimal surrounding subcutaneous edema.

I: Second-space intermetatarsal bursitis without associated Morton neuroma.

Case 4 — Sesamoiditis

Clinical: 27-year-old ballet dancer, focal pain under the first MTP on push-off, tender medial sesamoid.

F: The **medial (tibial) sesamoid** demonstrates diffuse marrow edema (low T1, high fat-suppressed T2) without a discrete fracture line or cortical step-off. There is mild surrounding soft-tissue edema and a trace first MTP effusion. The articular surface of the sesamoid and the **plantar plate** are intact. The lateral (fibular) sesamoid is normal. No fragmentation or avascular collapse.

I: Tibial sesamoiditis (marrow edema) without fracture or osteonecrosis.

Case 5 — Sesamoid fracture versus bipartite sesamoid

Clinical: 22-year-old male sprinter, acute pain after forced dorsiflexion of the great toe.

F: The **medial sesamoid** shows a transverse lucent line with irregular, sharply angulated margins and surrounding marrow edema on fat-suppressed PD; the two fragments together exceed the expected size of a single sesamoid, and there is bone marrow edema bridging the cleft. These features favor an acute fracture over a bipartite sesamoid, which typically shows smooth corticated margins without edema. Mild adjacent plantar soft-tissue edema. The plantar plate is intact.

I: Acute tibial sesamoid fracture; edema and irregular margins distinguish it from a bipartite variant.

Case 6 — Turf toe (first MTP capsuloligamentous injury)

Clinical: 24-year-old football lineman, hyperdorsiflexion of the great toe on artificial turf, painful push-off.

F: There is a partial-thickness tear of the **plantar plate** of the first MTP at the sesamoid-phalangeal junction with high T2 signal and a 2 mm interstitial gap. Edema surrounds the **medial collateral ligament** and there is proximal migration of the sesamoid complex by 3 mm relative to the contralateral expected position. A small first MTP effusion and bone marrow edema along the plantar first metatarsal head are present. The flexor hallucis longus is intact.

I: Grade II turf toe with partial plantar plate tear and sesamoid retraction.

Case 7 — First MTP osteoarthritis / hallux rigidus

Clinical: 61-year-old man, stiff painful great toe with limited dorsiflexion and a dorsal bump.

F: The **first MTP joint** shows asymmetric dorsal joint-space narrowing, dorsal osteophytes off the metatarsal head and proximal phalanx, and subchondral cysts up to 4 mm with subchondral marrow edema. There is a dorsal cartilage defect and a 5 mm intra-articular osteophyte/loose body dorsally. A small effusion is present. No erosive features or aggressive marrow replacement.

I: Hallux rigidus with dorsal-predominant first MTP osteoarthritis and dorsal loose body.

Case 8 — Freiberg infraction

Clinical: 16-year-old female gymnast, chronic pain and swelling at the second MTP.

F: The **second metatarsal head** is flattened with dorsal articular surface collapse, subchondral low T1 signal and patchy fat-suppressed T2 marrow edema, and a subchondral fracture line. There is mild joint-space widening with a small effusion and a 3 mm intra-articular fragment. Adjacent synovitis is present. The plantar plate is intact.

I: Freiberg infraction of the second metatarsal head with articular flattening and subchondral collapse.

Case 9 — Lisfranc ligament injury with diastasis

Clinical: 31-year-old man, midfoot pain after a fall from a curb with the foot plantarflexed; plantar ecchymosis.

F: The **Lisfranc ligament** (dorsal and interosseous components between the medial cuneiform and second metatarsal base) is disrupted with high T2 signal and discontinuity of the interosseous band. There is widening of the **first–second intermetatarsal interval** to 4 mm with malalignment at the second tarsometatarsal joint. Bone marrow edema involves the second metatarsal base and medial cuneiform, with a small avulsion fragment ("fleck"). Adjacent tarsometatarsal joint effusion and soft-tissue edema.

I: Lisfranc ligamentous injury with C1–M2 diastasis and associated osseous edema; unstable midfoot injury.

Case 10 — Second metatarsal stress fracture

Clinical: 29-year-old female marathon trainee with increasing mileage, focal dorsal forefoot pain.

F: The **second metatarsal diaphysis** demonstrates a transverse low-signal fracture line on T1 with surrounding circumferential marrow edema and periosteal edema/early callus on fat-suppressed PD. There is mild adjacent dorsal soft-tissue edema. No displacement or cortical break extending fully across. The remaining metatarsals are normal.

I: Second metatarsal diaphyseal stress fracture (high-grade marrow edema with a fracture line).

Case 11 — Navicular stress fracture

Clinical: 23-year-old male basketball player, vague dorsal midfoot pain over the "N spot."

F: The **tarsal navicular** shows a sagittally oriented stress fracture line in the central third, extending from the proximal articular surface, with surrounding marrow edema bright on fat-suppressed T2 and low on T1. No displacement or sclerotic nonunion margins. Mild talonavicular joint effusion. The posterior tibial tendon insertion is intact.

I: Central-third navicular stress fracture, the high-risk pattern warranting protected/non-weightbearing management.

Case 12 — Jones fracture (fifth metatarsal base)

Clinical: 20-year-old male soccer player, lateral foot pain after an inversion injury during a cut.

F: There is a transverse fracture at the **metaphyseal-diaphyseal junction of the fifth metatarsal** (Zone 2, Jones), approximately 1.8 cm distal to the styloid tip, with a low-signal fracture line and surrounding marrow edema. Minimal medial cortical step-off (<1 mm). No sclerosis to suggest chronicity. The **peroneus brevis** insertion and lateral band of the plantar fascia are intact.

I: Acute Jones fracture of the fifth metatarsal base (Zone 2); watershed location with delayed-union risk.

Case 13 — Midfoot osteoarthritis

Clinical: 64-year-old woman, chronic dorsal midfoot pain and a palpable dorsal ridge.

F: The **second and third tarsometatarsal joints** show joint-space narrowing, dorsal marginal osteophytes, subchondral cysts, and subchondral marrow edema. A dorsal **ganglion/synovial cyst** measuring 9 mm arises from the second TMT joint. No acute fracture or aggressive marrow signal. Mild surrounding soft-tissue edema.

I: Tarsometatarsal (Lisfranc complex) osteoarthritis with a dorsal ganglion.

Case 14 — Accessory navicular syndrome (type II)

Clinical: 17-year-old female, medial midfoot pain over a bony prominence aggravated by activity.

F: A **type II accessory navicular** is present, connected to the navicular tuberosity by a synchondrosis. There is marrow edema on both sides of the synchondrosis (accessory ossicle and navicular tuberosity) bright on fat-suppressed PD, indicating a symptomatic/stressed synchondrosis. The distal **posterior tibial tendon** is thickened with mild intrasubstance signal at its insertion but intact. No discrete fracture.

I: Symptomatic type II accessory navicular with synchondrosis edema and posterior tibial tendinosis.

Hindfoot & Plantar

Case 15 — Plantar fasciitis

Clinical: 48-year-old runner, insidious inferior heel pain worst with first steps in the morning.

F: The **proximal plantar fascia** at the calcaneal origin is thickened to 7 mm (normal <4 mm) with intrasubstance and perifascial fat-suppressed T2 hyperintensity. Mild reactive marrow edema in the inferior **calcaneal tuberosity** at the enthesis. No fascial discontinuity. A small plantar calcaneal enthesophyte is present.

I: Proximal plantar fasciitis with fascial thickening and enthesopathic marrow edema; no tear.

Case 16 — Plantar fascia tear

Clinical: 41-year-old male tennis player, sudden midfoot/arch pain with a pop during a lunge.

F: There is a partial-thickness tear of the **plantar fascia** just distal to the calcaneal origin, with focal fiber discontinuity, a 6 mm fluid-signal gap, and surrounding perifascial edema and hemorrhage. The fascia is thickened to 9 mm adjacent to the defect. No retraction of a complete tear; the central cord remains partially continuous. Adjacent plantar muscle edema.

I: Partial-thickness proximal plantar fascia tear with perifascial edema, superimposed on fasciitis.

Case 17 — Plantar fibromatosis (Ledderhose)

Clinical: 52-year-old man, palpable nontender nodules along the arch.

F: Along the **plantar aponeurosis** (medial cord), there are two fusiform nodular masses measuring 14 mm and 8 mm, contiguous with and arising from the superficial fascia, blending into the plantar fat. They are low-to-intermediate signal on T1 and predominantly low signal on T2 with mild heterogeneous enhancement. No infiltration of plantar muscles or bone. No fluid.

I: Plantar fibromatosis (Ledderhose disease) with characteristic low-T2 fascial nodules.

Case 18 — Calcaneal stress fracture

Clinical: 35-year-old female military recruit, diffuse heel pain with positive calcaneal squeeze.

F: Within the **posterior calcaneal body**, there is a vertically oriented serpiginous low-signal fracture line perpendicular to the trabeculae, with surrounding ill-defined marrow edema bright on STIR/fat-suppressed PD and low on T1. No cortical disruption or displacement. The plantar fascia and Achilles enthesis are normal.

I: Calcaneal body insufficiency/stress fracture with characteristic perpendicular line and marrow edema.

Case 19 — Peroneus longus tear / painful os peroneum syndrome

Clinical: 50-year-old man, lateral foot/cuboid pain after an inversion injury.

F: The **peroneus longus tendon** shows longitudinal intrasubstance high signal and partial-thickness splitting at the level of the cuboid tunnel. The **os peroneum** is fragmented with diastasis between fragments and surrounding tenosynovial fluid, consistent with painful os peroneum syndrome. There is mild marrow edema along the inferolateral cuboid at the peroneal groove. The peroneus brevis is intact.

I: Peroneus longus tendinopathy with partial tear and fragmented os peroneum (POPS).

Case 20 — Achilles insertional tendinosis with Haglund deformity

Clinical: 55-year-old recreational runner, posterior heel pain and swelling at the tendon insertion.

F: The distal **Achilles tendon** at its calcaneal insertion is thickened to 11 mm with intrasubstance intermediate signal and insertional enthesophyte formation. There is a prominent posterosuperior calcaneal process (**Haglund deformity**) with reactive retrocalcaneal bursitis (fluid >6 mm) and adjacent calcaneal marrow edema. No full-thickness tear; fibers remain continuous.

I: Insertional Achilles tendinosis with Haglund deformity and retrocalcaneal bursitis.

Other / Infection / Marrow

Case 21 — Osteomyelitis with diabetic foot ulcer

Clinical: 63-year-old diabetic man, plantar forefoot ulcer over a callus with surrounding erythema.

F: Beneath a plantar soft-tissue ulcer at the **first metatarsal head**, there is confluent low T1 marrow signal replacing the normal fatty marrow, with corresponding fat-suppressed T2 hyperintensity and avid post-contrast enhancement, plus cortical destruction. A sinus tract extends from the skin ulcer to the bone. Surrounding soft-tissue edema, phlegmon, and a small rim-enhancing collection are present. "Ghost sign" of the metatarsal head on T1.

I: First metatarsal head osteomyelitis contiguous with a plantar ulcer and adjacent soft-tissue infection.

Case 22 — Charcot neuroarthropathy

Clinical: 58-year-old diabetic woman with peripheral neuropathy, a swollen warm "rocker-bottom" midfoot without ulcer.

F: The **tarsometatarsal (Lisfranc) and midtarsal joints** show fragmentation, subluxation, and bony debris with periarticular marrow edema and large joint effusions, producing a rocker-bottom deformity. Marrow signal abnormality is centered on the joints (subchondral, periarticular) rather than subcortical with an overlying ulcer, and there is no sinus tract. Subchondral cysts and ligamentous disruption are present. No drainable abscess.

I: Midfoot Charcot neuroarthropathy; periarticular distribution and absence of an ulcer/sinus tract favor Charcot over osteomyelitis.

Case 23 — Soft-tissue abscess with cellulitis

Clinical: 44-year-old man, painful swollen dorsal foot after a puncture wound, febrile.

F: In the **dorsal midfoot subcutaneous and intermuscular soft tissues**, there is a 3.2 × 2.0 cm rim-enhancing fluid collection with central T2 hyperintensity and restricted diffusion, consistent with an abscess. Surrounding reticulated subcutaneous fat-suppressed T2 hyperintensity and skin thickening indicate cellulitis. No definite osseous marrow signal abnormality or cortical destruction to suggest osteomyelitis. No gas locules.

I: Dorsal foot soft-tissue abscess with overlying cellulitis; no MRI evidence of osteomyelitis.

Case 24 — Ganglion cyst

Clinical: 33-year-old woman, painless soft dorsal midfoot lump.

F: Arising adjacent to the **sinus tarsi/talonavicular region**, there is a well-circumscribed lobulated cystic lesion measuring 18 × 10 mm, homogeneously T2 hyperintense, T1 hypointense, with a thin neck/pedicle to the talonavicular joint and only thin peripheral enhancement. No internal solid nodularity or septal enhancement. No bone erosion.

I: Dorsal midfoot ganglion cyst arising from the talonavicular joint.

Case 25 — Tarsal coalition (talocalcaneal)

Clinical: 15-year-old male, recurrent ankle sprains, rigid flatfoot and hindfoot stiffness.

F: At the **middle subtalar facet**, there is a fibro-osseous **talocalcaneal coalition** with narrowing and irregularity of the facet, subchondral marrow edema on both sides, and a downsloping ("drunken waiter") configuration of the sustentaculum. No complete osseous bridge, indicating fibrocartilaginous coalition. Secondary findings include talar beaking. Posterior subtalar joint preserved.

I: Middle-facet talocalcaneal coalition (fibro-osseous) with reactive marrow edema.

Case 26 — Bone contusion / talar dome

Clinical: 26-year-old woman, acute inversion ankle injury, persistent deep hindfoot pain.

F: The **medial talar dome** shows geographic subchondral marrow edema (low T1, high fat-suppressed T2) without a discrete fracture line, cortical impaction, or osteochondral fragment. The overlying articular cartilage is intact. There is associated lateral ligament sprain (anterior talofibular ligament edema) and a small ankle effusion. No subchondral collapse.

I: Medial talar dome bone contusion from an inversion injury; no osteochondral fracture.

Case 27 — Avulsion fracture, fifth metatarsal styloid (pseudo-Jones)

Clinical: 38-year-old woman, lateral foot pain after an inversion injury stepping off a curb.

F: There is an avulsion fracture at the **tip of the fifth metatarsal styloid (Zone 1)** at the peroneus brevis/lateral plantar fascia attachment, with a minimally displaced fragment and surrounding marrow edema. The fracture line does not extend into the fourth–fifth intermetatarsal articulation. The **peroneus brevis** tendon is intact and taut. No Lisfranc injury.

I: Zone 1 avulsion ("pseudo-Jones") fracture of the fifth metatarsal styloid; favorable healing prognosis.

Case 28 — Gout (first MTP)

Clinical: 57-year-old man, recurrent acutely painful red swollen great toe, hyperuricemia.

F: At the **first MTP joint**, there is a juxta-articular soft-tissue mass (tophus) measuring 12 mm that is intermediate-to-low signal on T1 and heterogeneous intermediate signal on T2 with surrounding enhancement. There is a "punched-out" para-articular erosion with overhanging sclerotic margins along the medial metatarsal head, mild adjacent marrow edema, and a joint effusion. Relative preservation of joint space.

I: Tophaceous gout of the first MTP with characteristic erosions and a soft-tissue tophus.

Case 29 — Tibialis posterior tendon dysfunction with acquired flatfoot

Clinical: 59-year-old woman, progressive medial hindfoot pain and a flattening arch, "too many toes" sign.

F: The **posterior tibial tendon** at the level of the medial malleolus and navicular insertion is markedly thickened with longitudinal intrasubstance splitting and surrounding tenosynovial fluid; a focal high-grade partial tear is present. There is attenuation of the **spring (calcaneonavicular) ligament** with intermediate signal, hindfoot valgus, and talonavicular uncoverage consistent with acquired flatfoot. Sinus tarsi fat is replaced by edema/fibrosis.

I: Posterior tibial tendon dysfunction (Johnson stage II) with spring ligament insufficiency and pes planovalgus.

Case 30 — Osteochondral lesion of the talus

Clinical: 30-year-old man, chronic ankle pain and intermittent locking after a remote sprain.

F: The **medial talar dome** demonstrates an osteochondral lesion measuring 9 × 7 mm with a subchondral fracture and overlying cartilage fissuring. High fat-suppressed T2 signal at the fragment-bone interface indicates an unstable lesion; small subchondral cysts (up to 4 mm) are present. No fully displaced intra-articular body, though there is a small ankle effusion. Surrounding marrow edema.

I: Unstable medial talar dome osteochondral lesion with cystic change and high-signal interface; surgically relevant instability.